ErthaLoka Planetary Performance System (PPS)

Complete Technical Whitepaper

Di-MRVM Framework for Natural Capital Assessment

Version 2.0 | December 2024

Table of Contents

1. Executive Summary
2. PPS Framework Overview
3. ABCDE Pillar Definitions
4. Mathematical Framework
5. Complete Metric Parameters
6. Normalization Methodology
7. Scoring & Grading System
8. Ecosystem-Adaptive Weighting
9. Data Quality Score (DQS)
10. Economic Valuation
11. Data Sources
12. System Architecture
13. Technology Stack
14. API Specifications
15. Sources & References

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1. Executive Summary

The **Planetary Performance System (PPS)** is ErthaLoka's flagship environmental intelligence platform, designed to provide standardized, quantifiable assessments of natural capital and ecosystem health at any location on Earth.

Key Capabilities

Capability	Description
Real-time Assessment	Query any coordinates for instant environmental analysis
24 Environmental Indicators	Comprehensive metrics across 5 pillars
AAA-CCC Grading	Credit rating-style scores for easy interpretation
Economic Valuation	Ecosystem Service Value (ESV) in USD/ha/year
Carbon Quantification	IPCC Tier 1 carbon credit estimation
Data Quality Scoring	Confidence-weighted quality assessment

Target Users

- **Development Finance Institutions (DFIs)**: ESG due diligence for project financing
- **Commercial Banks**: Nature-related risk assessment for lending portfolios
- **Corporations**: Natural capital accounting and sustainability reporting
- **Governments**: Land use planning and environmental monitoring
- **Conservation Organizations**: Biodiversity and ecosystem tracking

Core Value Proposition

PPS transforms complex satellite data into actionable financial metrics, enabling:

- Nature-positive investment decisions
- Standardized environmental benchmarking
- Transparent, auditable ecosystem assessments
- Integration with existing ESG frameworks

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2. PPS Framework Overview

Di-MRVM: Digital Measure, Report, Verify, Monitor

PPS operates as a **Di-MRVM** (Digital Measurement, Reporting, Verification, and Monitoring) system, providing continuous environmental intelligence through satellite-derived data.

Framework Components

Component	Function	Output
Measure	Collect 24 environmental indicators via satellite	Raw metric values
Report	Normalize and aggregate into PPS score	AAA-CCC grade
Verify	Data Quality Score (DQS) assessment	Confidence level
Monitor	Track changes over time	Trend analysis

ABCDE + Q + R Framework

The PPS assessment framework consists of 5 core pillars plus 2 overlay dimensions:

Core Pillars (ABCDE)

Pillar	Name	Focus Area
A	Atmospheric Health	Air quality, aerosols, UV radiation
B	Biodiversity	Vegetation health, land cover, species habitat
C	Climate	Forest cover, biomass, carbon sequestration
D	Degradation Decrease	Soil health, water stress, drought resilience
E	Ecosystem Services	Human impact, ecosystem integrity

Overlay Dimensions

Overlay	Name	Function
Q	Quality	Data Quality Score (DQS) for confidence assessment
R	Risk/Resilience	Temporal trends and vulnerability indicators

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3. ABCDE Pillar Definitions

Pillar A: Atmospheric Health

Purpose: Assess air quality and atmospheric conditions affecting human and ecosystem health.

Metric	Description	Unit	Source
Aerosol Optical Depth (AOD)	Atmospheric particle concentration	dimensionless	MODIS
Air Quality Index (AQI)	Combined air pollution measure	US EPA scale	MODIS/External
UV Index	Ultraviolet radiation intensity	index (0-11+)	MODIS
Visibility	Atmospheric clarity	km	Derived
Cloud Fraction	Cloud cover percentage	fraction (0-1)	MODIS

Interpretation: Lower AOD and AQI values indicate cleaner air. UV Index uses sigmoid normalization for optimal mid-range values.

Pillar B: Biodiversity

Purpose: Evaluate vegetation health, ecosystem diversity, and habitat quality.

Metric	Description	Unit	Source
NDVI	Normalized Difference Vegetation Index	index (-1 to 1)	MODIS
EVI	Enhanced Vegetation Index	index (-1 to 1)	MODIS
LAI	Leaf Area Index	m^2/m^2	MODIS
FPAR	Fraction of Absorbed PAR	fraction (0-1)	MODIS
Land Cover	ESA WorldCover classification	class (10-100)	Sentinel-2

Interpretation: Higher NDVI/EVI indicates healthier, denser vegetation. NDVI > 0.6 typically represents dense forest or cropland.

Pillar C: Climate

Purpose: Quantify carbon storage capacity and forest ecosystem health. Carbon metrics are key components of this pillar.

Metric	Description	Unit	Source
Tree Cover	Percentage canopy cover	% (0-100)	Hansen GFC
Forest Loss	Cumulative loss since 2000	%	Hansen GFC
Canopy Height	Average tree height	meters	GEDI/ETH
Biomass	Above-ground biomass density	Mg/ha	GEDI
Carbon Stock	Estimated carbon storage	Mg C/ha	Derived

Interpretation: Higher tree cover and biomass indicate greater carbon sequestration capacity. Carbon stock is derived as 50% of biomass.

Pillar D: Decrease in Land & Water Degradation

Purpose: Monitor soil health, water availability, and degradation indicators.

Metric	Description	Unit	Source
Land Surface Temperature	Daytime surface temperature	°C	MODIS
Soil Moisture	Volumetric water content	m^3/m^3	SMAP/ERA5
Drought Index	Standardized drought measure	index (-3 to 3)	Derived
Evaporative Stress	Water-energy balance indicator	index (-2 to 2)	MODIS
Water Occurrence	Historical water presence	% (0-100)	JRC

Interpretation: Optimal soil moisture around 30%. Drought index near zero indicates normal conditions; negative values indicate drought.

Pillar E: Ecosystem Services

Purpose: Assess human impact and ecosystem service delivery capacity.

Metric	Description	Unit	Source
Human Modification Index	Degree of anthropogenic change	index (0-1)	CSP
Population Density	People per unit area	people/km^2	WorldPop
Nighttime Lights	Artificial light intensity	$nW/cm^2/sr$	VIIRS
Elevation	Terrain altitude	meters	SRTM
Distance to Water	Proximity to water bodies	km	Derived

Interpretation: Lower human modification indicates more pristine ecosystems. This pillar balances natural integrity against human presence.

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4. Mathematical Framework

4.1 Overall PPS Score Calculation

The Planetary Performance Score is calculated as a weighted sum of normalized pillar scores:

```
PPS = W_A × S_A + W_B × S_B + W_C × S_C + W_D × S_D + W_E × S_E
```

Where:

- S_X = Normalized score for pillar X (0-100)
- W_X = Ecosystem-adaptive weight for pillar X
- ΣW = 1.0 (weights sum to 100%)

4.2 Pillar Score Calculation

Each pillar score is computed from its constituent metrics:

```
S_pillar = Σ(w_i × N_i) / Σw_i
```

Where:

- w_i = Weight assigned to metric i within the pillar
- N_i = Normalized value of metric i (0-100)

- Only metrics with valid data are included

4.3 PHI-ESV Multiplier

The ESV adjustment multiplier based on PPS score:

```
M_PHI = [(PPS − 50) / 100] × k × [1 + α × ln(PPS / 50)]
```

Where:

- PPS = Overall Planetary Performance Score (0-100)
- k = 0.6 (base sensitivity factor)
- α = 0.15 (logarithmic acceleration factor)

Result Range: -0.30 (PPS=0) to +0.36 (PPS=100)

4.4 Carbon Credit Calculation

Based on IPCC Tier 1 methodology:

```
Carbon Stock (Mg C/ha) = Biomass × 0.5

CO₂ Equivalent (tonnes) = Carbon Stock × Area (ha) × 3.67

Verified Credits = CO₂ Equivalent × Confidence Factor
```

Where:

- 0.5 = Carbon fraction of biomass
- 3.67 = Molecular weight ratio (CO_2/C = 44/12)
- Confidence Factor = DQS / 100

4.5 Data Quality Score (DQS)

Criticality-weighted data quality assessment:

```
DQS = [Σ(w_i × a_i × q_i) / Σw_i] × 100
```

Where:

- w_i = Criticality weight for metric i
- a_i = Availability (1 if present, 0 if missing)
- q_i = Quality factor (good=1.0, moderate=0.8, poor=0.5)

4.6 Polygon Land Parcel Calculation

PPS supports polygon-based queries for land parcel analysis using 4 corner points. This enables precise ESV and carbon credit calculations for specific land areas.

4.6.1 Input: Four Corner Points

A polygon query requires exactly 4 geographic coordinates defining the land parcel corners:

Point	Position	Format
Point 1	Northwest (NW)	{lat: number, lng: number}

Point 2	Northeast (NE)	{lat: number, lng: number}
Point 3	Southeast (SE)	{lat: number, lng: number}
Point 4	Southwest (SW)	{lat: number, lng: number}

Constraints:

- Latitude: -90° to +90°
- Longitude: -180° to +180°
- Points should form a valid, non-self-intersecting polygon

4.6.2 Polygon Geometry Construction

Step 1: Create Closed Polygon

The 4 points are connected in order, with the first point repeated to close the polygon:

```
Polygon Coordinates = [P1, P2, P3, P4, P1]

Where each P = [longitude, latitude]  (GEE format)
```

Step 2: Calculate Centroid

The centroid serves as the reference point for the polygon:

```
Centroid_lat = (lat₁ + lat₂ + lat₃ + lat₄) / 4
Centroid_lng = (lng₁ + lng₂ + lng₃ + lng₄) / 4
```

4.6.3 Area Calculation

Method: Geodesic Area via Earth Engine

For accurate area calculation on Earth's curved surface:

```
Area (m²) = ee.Geometry.Polygon(coordinates).area().getInfo()
Area (hectares) = Area (m²) / 10,000
Area (acres) = Area (hectares) × 2.47105
```

Approximate Method (Shoelace Formula)

For quick estimation using planar approximation:

```
Area (degrees²) = |Σ(xᵢ × yᵢ₊₁ - xᵢ₊₁ × yᵢ)| / 2

Converting to hectares:
- Latitude scale: 111.32 km per degree
- Longitude scale: 111.32 × cos(avg_latitude) km per degree
- Area (km²) = Area (degrees²) × lat_scale × lng_scale
- Area (hectares) = Area (km²) × 100
```

4.6.4 Polygon Query Processing

Step 1: Geometry Validation

- Verify exactly 4 points provided
- Validate all coordinates within bounds
- Check polygon is non-self-intersecting

Step 2: Satellite Data Extraction

- Create Earth Engine polygon geometry
- Query all 5 pillars over the polygon region
- Use `reduceRegion` with mean reducer for spatial averaging

Step 3: Metric Aggregation

- Extract mean values for each metric within polygon
- Apply same normalization as point queries
- Calculate pillar scores using polygon-averaged values

Step 4: Area-Based Calculations

- Calculate total area in hectares
- Compute ESV based on area and ecosystem type
- Estimate carbon credits using biomass and area

4.6.5 ESV Calculation for Polygon

```
Total Annual ESV = Adjusted ESV per hectare × Area (hectares)

Where:
- Adjusted ESV = Base ESV × (1 + M_PHI)
- Base ESV = Ecosystem-specific value ($/ha/year)
- M_PHI = PPS-based multiplier
```

Projection Calculations:

```
10-Year ESV = Σ (Annual ESV × 1.02^i) for i = 0 to 9
30-Year ESV = Σ (Annual ESV × 1.02^i) for i = 0 to 29

Where 1.02 = 2% annual appreciation factor
```

4.6.6 Carbon Credits for Polygon

```
Total Carbon Stock (Mg C) = Carbon Stock (Mg C/ha) × Area (hectares)

Total CO₂ Equivalent = Total Carbon Stock × 3.67

Verified Credits = Total CO₂ × Confidence Factor

Confidence Factor = DQS / 100
```

Value Estimation:

```
Low Value  = Verified Credits × $15/tonne
Mid Value  = Verified Credits × $25/tonne
High Value = Verified Credits × $50/tonne
```

4.6.7 Polygon Query Response Structure

Field	Description
geometry.type	"Polygon"

geometry.points	Array of 4 input points
geometry.centroid	Calculated center point
geometry.area_m2	Area in square meters
geometry.area_ha	Area in hectares
geometry.area_acres	Area in acres
carbon_credits.co2_equivalent_tonnes	Total CO_2 equivalent
carbon_credits.verified_co2_tonnes	DQS-adjusted credits
carbon_credits.estimated_value	Low/mid/high USD values
ecosystem_service_value.total_annual_esv_usd	Annual ESV for parcel
ecosystem_service_value.projections	10-year and 30-year values

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4.6.8 Example: 100-Hectare Forest Parcel

Input Points:

Point	Latitude	Longitude
NW	28.6200	77.2000
NE	28.6200	77.2200
SE	28.6000	77.2200
SW	28.6000	77.2000

Calculated Results:

Parameter	Value
Area	100 hectares
Ecosystem Type	Tropical Forest
PPS Score	72 (Grade AA)
DQS	85%
Biomass	200 Mg/ha
Carbon Stock	100 Mg C/ha
Total Carbon	10,000 Mg C
CO_2 Equivalent	36,700 tonnes
Verified Credits	31,195 tonnes
Carbon Value (Mid)	$779,875
Base ESV	$5,382/ha/year
PPS Multiplier	+0.15

Adjusted ESV	$6,189/ha/year
Total Annual ESV	$618,900/year
10-Year ESV	$6,783,542
30-Year ESV	$25,043,789

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5. Complete Metric Parameters

5.1 Category A: Atmospheric Health

Metric	v_min	v_max	v_mid/v_opt	Normalization	Weight	Criticality
aod	0.0	1.5	-	Inverse Linear	0.40	Important
aqi	0	300	-	Inverse Linear	0.40	Important
uv_index	0	11	5.5	Sigmoid (k=0.5)	0.10	Auxiliary
visibility	1	50	-	Linear	0.10	Auxiliary
cloud_fraction	0.0	1.0	-	Inverse Linear	0.00	Auxiliary

5.2 Category B: Biodiversity

Metric	v_min	v_max	v_mid/v_opt	Normalization	Weight	Criticality
ndvi	-0.1	0.9	-	Linear	0.35	**Critical**
evi	0.0	1.0	-	Linear	0.35	Important
lai	0.0	7.0	3.5	Sigmoid (k=0.5)	0.15	Supporting
fpar	0.0	0.95	-	Linear	0.15	Supporting
land_cover	10	100	-	Linear	0.00	Auxiliary

5.3 Category C: Climate

Metric	v_min	v_max	v_mid/v_opt	Normalization	Weight	Criticality
tree_cover	0	100	-	Linear	0.35	**Critical**
biomass	0	400	200	Sigmoid (k=0.5)	0.35	Important
canopy_height	0	40	20	Sigmoid (k=0.5)	0.15	Important
forest_loss	0	10	-	Inverse Linear	0.15	Supporting
carbon_stock	0	200	100	Sigmoid (k=0.5)	0.00	Important

5.4 Category D: Degradation Decrease

Metric	v_min	v_max	v_opt/sigma	Normalization	Weight	Criticality

soil_moisture	0.05	0.45	0.30 (σ=0.12)	Gaussian	0.35	**Critical**
drought_index	-2.0	2.0	0	Centered	0.35	Important
lst	15	45	25 (σ=8)	Gaussian	0.15	Supporting
evaporative_stress	-1.0	1.0	0	Centered	0.15	Supporting
water_occurrence	0	100	-	Linear	0.00	Auxiliary

5.5 Category E: Ecosystem Services

Metric	v_min	v_max	v_mid/k	Normalization	Weight	Criticality
human_modification	0.0	1.0	-	Inverse Linear	0.50	**Critical**
population	0	5000	500	Inv. Sigmoid (k=0.3)	0.17	Supporting
nightlights	0	63	20	Inv. Sigmoid (k=0.3)	0.17	Supporting
distance_to_water	0	50	-	Inverse Linear	0.16	Auxiliary
elevation	0	5000	-	Linear	0.00	Auxiliary

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6. Normalization Methodology

6.1 Normalization Function Types

PPS uses 6 distinct normalization functions to transform raw metric values into standardized 0-100 scores:

Linear Normalization

Use Case: Metrics where higher values are always better (e.g., NDVI, tree cover)

```
Score = [(Value − v_min) / (v_max − v_min)] × 100
```

Metrics: ndvi, evi, fpar, tree_cover, visibility, water_occurrence

---

Inverse Linear Normalization

Use Case: Metrics where lower values are better (e.g., pollution, degradation)

```
Score = [1 − (Value − v_min) / (v_max − v_min)] × 100
```

Metrics: aod, aqi, forest_loss, human_modification, distance_to_water

---

Sigmoid Normalization

Use Case: Metrics with diminishing returns at high values

```
Score = [1 / (1 + e^(−k × (x − x_mid) × 10))] × 100

Where: x = (Value − v_min) / (v_max − v_min)
       x_mid = (v_mid − v_min) / (v_max − v_min)
```

Metrics: uv_index, lai, biomass, canopy_height, carbon_stock

Inverse Sigmoid Normalization

Use Case: Metrics where lower values are better with saturation effects

```
Score = [1 / (1 + e^(k × (x − x_mid) × 10))] × 100
```

Metrics: population, nightlights

Gaussian Normalization

Use Case: Metrics with an optimal value (not at extremes)

```
Score = e^[−0.5 × ((Value − v_opt) / σ)²] × 100
```

Metrics: lst (optimal ~25°C), soil_moisture (optimal ~0.30)

Centered Normalization

Use Case: Metrics where zero is optimal (deviation in either direction is negative)

```
Score = [1 − |Value| / max(|v_min|, |v_max|)] × 100
```

Metrics: drought_index, evaporative_stress

6.2 Normalization Selection Guide

Condition	Normalization Type
Higher is always better	Linear
Lower is always better	Inverse Linear
Optimal at specific value	Gaussian
Optimal at zero	Centered
Higher better with saturation	Sigmoid
Lower better with saturation	Inverse Sigmoid

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7. Scoring & Grading System

7.1 AAA-CCC Grading Scale

PPS uses a credit rating-style grading system for intuitive interpretation:

Grade	Score Range	Interpretation	Color Code
AAA	86 - 100	Excellent	#22c55e (Green)
AA	72 - 85	Very Good	#4ade80 (Light Green)
A	58 - 71	Good	#84cc16 (Lime)
BBB	44 - 57	Above Average	#facc15 (Yellow)

BB	30 - 43	Average	#f59e0b (Amber)
B	16 - 29	Below Average	#f97316 (Orange)
CCC	0 - 15	Poor	#ef4444 (Red)

7.2 Grade Interpretation Guide

Grade	Ecosystem Status	Investment Implication
AAA	Pristine, high-functioning ecosystem	Premium natural capital asset
AA	Healthy ecosystem with minor pressures	Strong investment candidate
A	Good condition, some management needed	Positive outlook
BBB	Moderate health, improvement potential	Investment grade threshold
BB	Average condition, notable pressures	Requires due diligence
B	Degraded, significant intervention needed	High-risk, high-reward potential
CCC	Severely degraded ecosystem	Restoration project candidate

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7.3 Comparison with Financial Ratings

PPS Grade	Equivalent Credit Rating	Risk Level
AAA	S&P AAA / Moody's Aaa	Minimal
AA	S&P AA / Moody's Aa	Very Low
A	S&P A / Moody's A	Low
BBB	S&P BBB / Moody's Baa	Moderate
BB	S&P BB / Moody's Ba	Elevated
B	S&P B / Moody's B	High
CCC	S&P CCC / Moody's Caa	Very High

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8. Ecosystem-Adaptive Weighting

8.1 Weight Distribution by Ecosystem Type

PPS automatically detects ecosystem type from land cover and adjusts pillar weights accordingly:

Ecosystem Type	A (Atm)	B (Bio)	C (Carbon)	D (Degrad)	E (Eco)	Priority Focus
Tropical Forest	0.10	0.25	**0.35**	0.15	0.15	Carbon storage
Mangrove	0.10	0.20	**0.30**	**0.25**	0.15	Carbon + water
Grassland/Savanna	0.15	**0.30**	0.15	**0.25**	0.15	Biodiversity
Wetland	0.10	0.25	0.20	**0.30**	0.15	Water conditions

Agricultural	0.20	0.25	0.10	0.25	0.20	Balanced
Urban Green	**0.30**	0.25	0.10	0.15	0.20	Air quality
Default/Mixed	0.20	0.20	0.20	0.20	0.20	Equal weights

8.2 Land Cover to Ecosystem Mapping

WorldCover Class	Code	Ecosystem Type Assigned
Tree cover	10	Tropical Forest
Shrubland	20	Grassland/Savanna
Grassland	30	Grassland/Savanna
Cropland	40	Agricultural
Built-up	50	Urban Green
Bare/sparse vegetation	60	Default
Snow and ice	70	Default
Permanent water	80	Wetland
Herbaceous wetland	90	Wetland
Mangroves	95	Mangrove
Moss and lichen	100	Default

8.3 Rationale for Adaptive Weighting

Ecosystem	Weight Emphasis	Scientific Rationale
Tropical Forest	Carbon (35%)	Highest carbon density ecosystems globally
Mangrove	Carbon + Degradation	Blue carbon + coastal vulnerability
Grassland	Biodiversity + Degradation	Species richness + fire/grazing pressure
Wetland	Degradation (30%)	Water-dependent, drought-sensitive
Agricultural	Balanced	Multiple ecosystem services provided
Urban Green	Atmospheric (30%)	Air quality critical for human health

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9. Data Quality Score (DQS)

9.1 Criticality Weights

Metrics are assigned criticality levels that determine their weight in DQS calculation:

Criticality Level	Weight	Metrics Included
Critical	1.0	ndvi, tree_cover, soil_moisture, human_modification

Important	0.7	aod, aqi, evi, biomass, canopy_height, drought_index, carbon_stock
Supporting	0.4	lai, fpar, forest_loss, lst, population, nightlights, evaporative_stress
Auxiliary	0.2	uv_index, visibility, cloud_fraction, land_cover, water_occurrence, elevation, distance_to_water

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9.2 Quality Factor Adjustments

Data Quality	Factor	Condition
Good	1.0	Fresh data, high resolution, no artifacts
Moderate	0.8	Slightly dated or lower resolution
Poor	0.5	Significant gaps or quality issues
Unavailable	0.0	No data available

9.3 DQS Threshold Interpretation

DQS Range	Level	Recommendation
85 - 100	High Confidence	Suitable for all applications
70 - 84	Investment Grade	Acceptable for financial decisions
50 - 69	Acceptable	Consider supplementing with ground data
40 - 49	Marginal	Interpret with caution
0 - 39	Low Confidence	Insufficient for decision-making

9.4 Missing Critical Metrics Impact

If critical metrics are unavailable, DQS is significantly impacted:

Missing Critical Metric	DQS Impact
NDVI	-25 points approx.
Tree Cover	-25 points approx.
Soil Moisture	-25 points approx.
Human Modification	-25 points approx.
All 4 critical metrics	DQS ≈ 30-40

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10. Economic Valuation

10.1 Ecosystem Service Value (ESV)

Base ESV by Ecosystem Type

Ecosystem Type	Base ESV ($/ha/year)	Primary Services

Wetland	$25,682	Water regulation, habitat, nutrient cycling
Mangrove	$9,990	Coastal protection, carbon, nursery habitat
Tropical Forest	$5,382	Carbon sequestration, biodiversity, water
Urban Green	$3,212	Air quality, recreation, temperature regulation
Grassland/Savanna	$2,871	Grazing, carbon, biodiversity
Agricultural	$1,532	Food provisioning, pollination
Default/Mixed	$3,000	Global average

Source: Costanza et al. (2014), de Groot et al. (2012)

ESV Adjustment Formula

```
Adjusted ESV = Base ESV × (1 + M_PHI)

Where M_PHI = [(PPS − 50) / 100] × 0.6 × [1 + 0.15 × ln(PPS / 50)]
```

ESV Multiplier Examples

PPS Score	Grade	Multiplier	Effect on ESV
90	AAA	+0.29	+29% premium
75	AA	+0.18	+18% premium
60	A	+0.07	+7% premium
50	BBB	0.00	Baseline
40	BB	-0.07	-7% discount
25	B	-0.18	-18% discount
10	CCC	-0.30	-30% discount

ESV Service Breakdown

Service Category	Allocation	Examples
Provisioning	20%	Food, water, timber, fiber
Regulating	35%	Climate, flood control, pollination
Cultural	15%	Recreation, aesthetic, spiritual
Supporting	30%	Nutrient cycling, soil formation

10.2 Carbon Credit Valuation

Calculation Methodology

Step 1: Carbon Stock Estimation

```
Carbon Stock (Mg C/ha) = Above-ground Biomass × 0.5
```

Step 2: Total Carbon

```
Total Carbon (Mg C) = Carbon Stock × Area (hectares)
```

Step 3: CO_2 Equivalent

```
CO₂ Equivalent (tonnes) = Total Carbon × 3.67
```

Step 4: Verified Credits

```
Verified Credits = CO₂ Equivalent × (DQS / 100)
```

Carbon Price Scenarios

Price Scenario	$/tonne CO_2	Market Reference
Low	$15	Voluntary market floor
Mid	$25	Average voluntary credit
High	$50	Premium/Compliance markets

Example Calculation

For a 100-hectare tropical forest with:

- Biomass: 250 Mg/ha
- DQS: 80%

Step	Calculation	Result
Carbon Stock	250 × 0.5	125 Mg C/ha
Total Carbon	125 × 100	12,500 Mg C
CO_2 Equivalent	12,500 × 3.67	45,875 tonnes CO_2
Verified Credits	45,875 × 0.80	36,700 tCO_2
Value (Mid)	36,700 × $25	**$917,500**

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11. Data Sources

11.1 Primary Satellite Constellations

Constellation	Agency	Sensors	Resolution	Revisit	Primary Use
MODIS	NASA	Terra/Aqua	250m-1km	Daily	Vegetation, atmosphere, temperature
Sentinel-2	ESA	MSI	10-60m	5 days	High-res optical, land cover
Landsat 8/9	USGS	OLI/TIRS	30m	16 days	Historical analysis, thermal
GEDI	NASA	LiDAR	25m	-	Canopy height, biomass

SMAP	NASA	Radar	9km	2-3 days	Soil moisture
VIIRS	NOAA	DNB	500m	Daily	Nighttime lights

11.2 Derived Products

Product	Provider	Resolution	Update	Metrics Derived
Hansen Global Forest Change	U. Maryland	30m	Annual	Tree cover, forest loss
ESA WorldCover	ESA	10m	Annual	Land cover classification
Global Human Modification	CSP	1km	Static	Human modification index
JRC Global Surface Water	JRC	30m	Monthly	Water occurrence
WorldPop	U. Southampton	100m	Annual	Population density
SRTM DEM	NASA/NGA	30m	Static	Elevation

11.3 External APIs (Supplementary Data)

API	Provider	Data Type	Update Frequency
Open-Meteo	Open-Meteo	Weather, soil moisture	Hourly
OpenAQ	OpenAQ	Air quality (ground stations)	Real-time

11.4 Complete GEE Dataset Catalog

Dataset ID	Metrics	Scale	Temporal
MODIS/061/MCD19A2_GRANULES	AOD	1km	Daily
MODIS/061/MOD13A2	NDVI, EVI	1km	16-day
MODIS/061/MOD15A2H	LAI, FPAR	500m	8-day
MODIS/061/MOD11A2	LST	1km	8-day
MODIS/061/MOD16A2	ET, PET	500m	8-day
ESA/WorldCover/v200	Land Cover	10m	Annual
UMD/hansen/global_forest_change_2023_v1_11	Tree Cover, Loss	30m	Annual
LARSE/GEDI/GEDI04_A_002_MONTHLY	Biomass	1km	Monthly
NASA/SMAP/SPL4SMGP/007	Soil Moisture	11km	3-hourly
CSP/HM/GlobalHumanModification	HMI	1km	Static
WorldPop/GP/100m/pop_age_sex_cons_unadj	Population	100m	Annual
NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG	Nightlights	500m	Monthly
JRC/GSW1_4/GlobalSurfaceWater	Water	30m	Static

USGS/SRTMGL1_003	Elevation	30m	Static

---

12. System Architecture

12.1 High-Level Architecture

```
┌──────────────────────────────────────────────────────────────┐
│                       CLIENT LAYER                           │
│   Web Application (React) + Mobile (Future)                  │
│   - Interactive Maps    - Report Generation                  │
│   - Data Visualization  - User Authentication                │
└──────────────────────────────┬───────────────────────────────┘
                               │
                        REST API (HTTPS)
                               │
┌──────────────────────────────▼───────────────────────────────┐
│                     API GATEWAY LAYER                        │
│   FastAPI Server                                             │
│   - Query Orchestration  - Request Validation                │
│   - Rate Limiting        - CORS Management                   │
└──────────────────────────────┬───────────────────────────────┘
                               │
      ┌──────────┬─────────────┼────────────┬──────────┐
      ▼          ▼             ▼            ▼          ▼
┌─────────┐ ┌─────────┐ ┌─────────┐ ┌─────────┐ ┌─────────┐
│Pillar A │ │Pillar B │ │Pillar C │ │Pillar D │ │Pillar E │
│ Atmos.  │ │ Biodiv. │ │ Carbon  │ │ Degrad. │ │ Ecosys. │
└────┬────┘ └────┬────┘ └────┬────┘ └────┬────┘ └────┬────┘
     └──────────┴──────────┴────────────┴──────────┘
                               │
┌──────────────────────────────▼───────────────────────────────┐
│                 DATA PROCESSING LAYER                        │
│   - Normalization Engine  - Scoring Aggregation              │
│   - DQS Calculation       - ESV/Carbon Computation           │
└──────────────────────────────┬───────────────────────────────┘
                               │
      ┌────────────────────────┼────────────────────────┐
      ▼                        ▼                        ▼
┌──────────────┐    ┌──────────────────┐    ┌──────────────────┐
│ Google Earth │    │  External APIs   │    │  IoT Sensors     │
│ Engine       │    │  (Weather, AQI)  │    │  (Real-time)     │
│ 50+ PB Data  │    │                  │    │                  │
└──────────────┘    └──────────────────┘    └──────────────────┘
```

12.2 Query Processing Flow

Step	Process	Duration
1	Request validation	<100ms
2	Geometry creation	<100ms
3	Parallel pillar queries (5)	10-25s

4	Metric extraction & normalization	<500ms
5	Score aggregation	<100ms
6	DQS calculation	<100ms
7	ESV/Carbon computation	<100ms
8	Response assembly	<100ms
Total	**End-to-end**	**12-30s**

12.3 Query Modes

Mode	Metrics Queried	Typical Duration	Use Case
Simple	10 core metrics	8-15 seconds	Quick assessment
Comprehensive	24 full metrics	20-35 seconds	Detailed analysis

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13. Technology Stack

13.1 Frontend Technologies

Category	Technology	Purpose
Framework	React 18	User interface
Language	TypeScript 5	Type safety
Build Tool	Vite	Fast development
Styling	TailwindCSS	Utility-first CSS
State Management	TanStack Query	Server state
Maps	Leaflet + React-Leaflet	Interactive mapping
Charts	Recharts + Chart.js	Data visualization
UI Components	Radix UI + shadcn/ui	Accessible components
Forms	React Hook Form + Zod	Validation
PDF Export	jsPDF + html2canvas	Report generation

13.2 Backend Technologies

Category	Technology	Purpose
Framework	FastAPI	REST API
Language	Python 3.10+	Server logic
Satellite Data	Google Earth Engine API	Geospatial processing
Validation	Pydantic	Request/response models

Data Processing	Pandas + NumPy	Numerical operations
Geospatial	GeoPandas + Shapely	Geometry handling
HTTP Client	aiohttp	Async API calls
Database	Supabase (PostgreSQL)	Persistent storage
Real-time DB	Firebase Firestore	Sensor data
Authentication	Firebase Auth	User management

13.3 Infrastructure

Component	Service	Purpose
Frontend Hosting	Vercel / Netlify	Static deployment
Backend Hosting	Cloud Run / Railway	Container deployment
Database	Supabase	PostgreSQL + Auth
Satellite Processing	Google Earth Engine	Cloud computing
CDN	CloudFlare	Content delivery
Monitoring	Cloud Logging	Observability

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14. API Specifications

14.1 REST API Endpoints

Method	Endpoint	Description	Auth Required
POST	`/api/query`	Point-based PPS query	Optional
POST	`/api/query/polygon`	Polygon-based query with ESV & Carbon	Optional
GET	`/api/health`	API health check	No
GET	`/api/datasets`	Available metrics list	No
GET	`/api/imagery`	Satellite imagery tiles	No
POST	`/api/pdf`	Generate PDF report	Yes
GET	`/api/history`	User query history	Yes
GET	`/api/external/weather`	Weather forecast	No
GET	`/api/external/air-quality`	Real-time AQI	No
GET	`/api/external/soil`	Soil moisture data	No

14.2 Point Query Request

Parameter	Type	Required	Description

latitude	float	Yes	Latitude (-90 to 90)
longitude	float	Yes	Longitude (-180 to 180)
mode	string	No	"simple" or "comprehensive" (default)
temporal	string	No	"latest", "monthly", "annual"
buffer_radius	int	No	Buffer in meters (100-10000, default 500)
pillars	array	No	Specific pillars ["A","B","C","D","E"]

14.3 Polygon Query Request

Parameter	Type	Required	Description
points	array	Yes	4 corner points [{lat, lng}, ...]
mode	string	No	"simple" or "comprehensive"
include_scores	bool	No	Include ESV and carbon (default true)

14.4 Response Structure

Field	Type	Description
query	object	Query metadata (coordinates, timestamp, mode)
pillars	object	Per-pillar data with metrics and scores
summary	object	Aggregated scores, DQS, ESV multiplier
time_series	object	Temporal mode information

14.5 Summary Response Fields

Field	Type	Description
overall_score	int	PPS score (0-100)
overall_interpretation	string	Grade (AAA-CCC)
pillar_scores	object	Individual pillar scores
ecosystem_type	string	Detected ecosystem
ecosystem_weights	object	Applied pillar weights
data_quality_score	float	DQS (0-100)
data_completeness	float	Percentage of available metrics
missing_critical_metrics	array	List of missing critical metrics
esv_multiplier	float	PHI-ESV adjustment factor

---

15. Sources & References

15.1 Scientific Literature

Ecosystem Service Valuation

1. **Costanza, R., de Groot, R., Sutton, P., et al. (2014)**
   - "Changes in the global value of ecosystem services"
   - *Global Environmental Change*, 26, 152-158
   - DOI: 10.1016/j.gloenvcha.2014.04.002
2. **de Groot, R., Brander, L., van der Ploeg, S., et al. (2012)**
   - "Global estimates of the value of ecosystems and their services in monetary units"
   - *Ecosystem Services*, 1(1), 50-61
   - DOI: 10.1016/j.ecoser.2012.07.005
3. **TEEB (2010)**
   - "The Economics of Ecosystems and Biodiversity: Mainstreaming the Economics of Nature"
   - UNEP, Geneva

Forest & Carbon

4. **Hansen, M.C., Potapov, P.V., Moore, R., et al. (2013)**
   - "High-Resolution Global Maps of 21st-Century Forest Cover Change"
   - *Science*, 342(6160), 850-853
   - DOI: 10.1126/science.1244693
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   - "2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories"
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6. **Dubayah, R., et al. (2020)**
   - "The Global Ecosystem Dynamics Investigation: High-resolution laser ranging of the Earth's forests and topography"
   - *Science of Remote Sensing*, 1, 100002

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7. **Vermote, E. (2015)**
   - "MOD09A1 MODIS/Terra Surface Reflectance 8-Day L3 Global 500m SIN Grid V006"
   - NASA EOSDIS Land Processes DAAC
   - DOI: 10.5067/MODIS/MOD09A1.006
8. **Drusch, M., Del Bello, U., Carlier, S., et al. (2012)**
   - "Sentinel-2: ESA's Optical High-Resolution Mission for GMES Operational Services"
   - *Remote Sensing of Environment*, 120, 25-36
9. **Farr, T.G., et al. (2007)**
   - "The Shuttle Radar Topography Mission"
   - *Reviews of Geophysics*, 45(2)

15.2 Data Source Documentation

Source	Documentation URL
Google Earth Engine	developers.google.com/earth-engine
MODIS Products	modis.gsfc.nasa.gov
Sentinel-2	sentinel.esa.int/web/sentinel/missions/sentinel-2
Hansen GFC	earthenginepartners.appspot.com/science-2013-global-forest
ESA WorldCover	esa-worldcover.org
GEDI	gedi.umd.edu

15.3 Standards & Frameworks

Standard	Organization	Application
US AQI	US EPA	Air Quality Index scale
IPCC Tier 1	IPCC	Carbon accounting methodology
Verra VCS	Verra	Verified Carbon Standard
Gold Standard	Gold Standard	Premium carbon credits
TNFD	TNFD	Nature-related financial disclosures
SBTN	SBTN	Science-based targets for nature

15.4 Technology Documentation

Technology	Documentation
FastAPI	fastapi.tiangolo.com
React	react.dev
TypeScript	typescriptlang.org
TailwindCSS	tailwindcss.com
Supabase	supabase.com/docs
Firebase	firebase.google.com/docs
Open-Meteo	open-meteo.com/en/docs
OpenAQ	docs.openaq.org

Appendix: Quick Reference Cards

A. Normalization Quick Reference

Function	Formula	Use When
Linear	(v - min) / (max - min)	Higher = better

Inverse Linear	1 - (v - min) / (max - min)	Lower = better
Sigmoid	1 / (1 + e^-x)	Diminishing returns
Gaussian	e^(-0.5 × z^2)	Optimal at specific value
Centered	1 - |v| / max	Optimal at zero

B. Grade Quick Reference

Grade	Range	Meaning
AAA	86-100	Excellent
AA	72-85	Very Good
A	58-71	Good
BBB	44-57	Above Average
BB	30-43	Average
B	16-29	Below Average
CCC	0-15	Poor

C. ESV Quick Reference

Ecosystem	Base ESV ($/ha/yr)
Wetland	$25,682
Mangrove	$9,990
Tropical Forest	$5,382
Urban Green	$3,212
Grassland	$2,871
Agricultural	$1,532

D. Critical Metrics Quick Reference

Criticality	Weight	Metrics
Critical (1.0)	Highest	NDVI, Tree Cover, Soil Moisture, Human Modification
Important (0.7)	High	AOD, AQI, EVI, Biomass, Drought Index
Supporting (0.4)	Medium	LAI, FPAR, LST, Population
Auxiliary (0.2)	Low	UV, Elevation, Cloud Cover

Document Version: 2.0 Published: December 2024 ErthaLoka Technologies

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This technical whitepaper is intended for institutional investors, development finance institutions, corporate sustainability teams, and environmental professionals seeking to understand the technical foundations of the ErthaLoka Planetary Performance System.